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Project Title	Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)
Week	Week 2

WEEK-2 TASK

Task Name

Dynamic Contextual Pricing Engine with Multi-Armed Bandits (Thompson Sampling)

Abstract

Dynamic Contextual Pricing Engine is an AI-based pricing system that automatically adjusts product prices based on customer behavior, demand, and market conditions. It uses the **Thompson Sampling** algorithm, a Multi-Armed Bandit approach, to balance exploration and exploitation while maximizing revenue. The system continuously learns from customer responses and recommends the most profitable price for each situation. This approach improves sales, customer satisfaction, and business profitability.

Introduction

Pricing is one of the most important factors in business success. Traditional pricing methods use fixed prices or manual updates, which may not adapt to changing customer preferences. A Dynamic Contextual Pricing Engine uses machine learning to optimize prices in real time. Thompson Sampling helps the system test different prices and gradually select the best-performing one based on customer purchase behavior.

Theory

A **Multi-Armed Bandit (MAB)** is a reinforcement learning problem where each pricing option is treated as an "arm" of a slot machine. The goal is to maximize total reward by choosing the best price.

Thompson Sampling is a Bayesian algorithm that:

- Learns from customer feedback.

- Explores new pricing options.
- Exploits the best-performing prices.
- Continuously updates pricing decisions.

Objectives

- Build an intelligent dynamic pricing system.
- Maximize revenue and profit.
- Learn customer purchasing behavior.
- Balance exploration and exploitation.
- Improve pricing decisions using AI.

Methodology

1. Collect customer and product data.
2. Define multiple pricing options.
3. Initialize Thompson Sampling model.
4. Display selected prices to customers.
5. Record purchase/no-purchase outcomes.
6. Update probability distributions.
7. Recommend the best price continuously.

Dataset

Example dataset fields:

Attribute	Description
Product_ID	Product identifier
Customer_ID	Customer identifier
Price	Offered price
Purchase	1 = Bought, 0 = Not Bought
Time	Purchase time
Device	Mobile/Desktop
Location	Customer location
Demand_Level	Low/Medium/High

Thompson Sampling Mechanics

1. Assign a probability distribution to each price.
2. Randomly sample from each distribution.
3. Select the price with the highest sampled value.
4. Observe customer response.
5. Update success/failure counts.
6. Repeat the process to improve pricing decisions.

Complexity Analysis

Operation	Complexity
Price Selection	$O(k)$
Distribution Update	$O(1)$
Total for N customers	$O(N \times k)$

Where:

- N = Number of customers
- k = Number of price options

Results

- Increased revenue through optimized pricing.
- Improved customer engagement.
- Better conversion rates.
- Faster learning of customer preferences.
- More accurate pricing recommendations over time.

Applications

- E-commerce websites
- Airline ticket pricing
- Hotel room pricing
- Ride-sharing services
- Online advertising
- Retail stores
- Food delivery platforms
- Subscription services

Conclusion

The **Dynamic Contextual Pricing Engine using Thompson Sampling** provides an effective AI-based solution for real-time pricing optimization. By balancing exploration and exploitation, it identifies the most profitable prices while adapting to customer behavior and market changes. This approach improves revenue, customer satisfaction, and decision-making, making it suitable for modern businesses that require intelligent and adaptive pricing strategies.